بسم الله الرحمن الرحيم

(DSP)

- Signal → any Physical quality which varies with time or any another independent variable.

Ex: $x_1(t) = 3t$, $x_2(t) = A\cos\omega t$.

- System → "hardware and software realization" It is a Physical device which Performe Some Perations on the signal

Ex: Filter i/p signal → LPF → o/p signal → Programming.

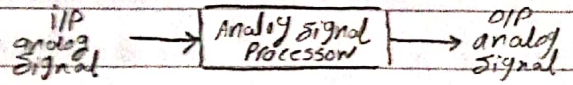
- Signal Processing → group of operations performe on the signal.

Ex: Filteration.

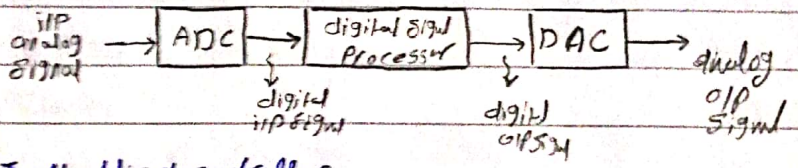
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*Types of Signal Processing

1) Analog Signal Processing (ASP)



2) Digital Signal Processing (DSP)



Ex: Small Microcontroller.

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* Advantage of DSP over ASP

1) high Flexability: reconfiguration operation simply by changing ~~program~~.

2) high accuracy

3) DSP implemintation is cheper than analog.

4) ~~easily~~ digital signal can easily stored.

---

* Classification of Signal.

1) Multidimension: related to number of independant variables.

Ex: $S_1(t) = 3t - 2$ "one dim", $S_2(t,x) = 3t + 8x$. "Multidim"

2) Multi channel: related to number of Sources

Ex: $S(t) = \begin{cases} S_1(t) \\ S_2(t) \\ S_3(t) \end{cases}$ → Multi. Channel

2) Continous time signal: defined for all ~~digital~~ time

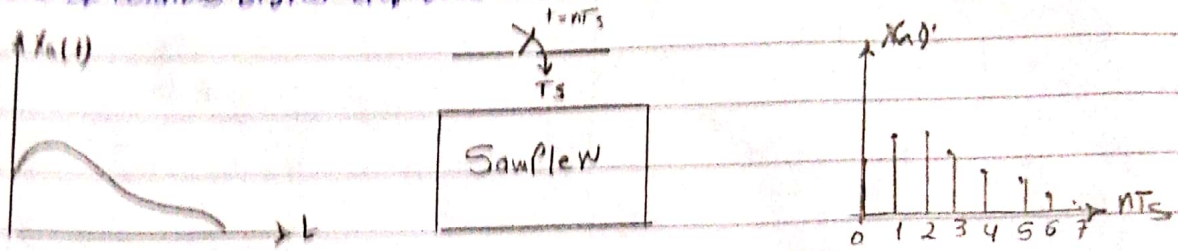
VS

Descrite time signal: defined for only certain time.

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State and explain Sampling theorem?

Sampling: Conversion of continuous time signal into discrete time signal by taking samples of continuous signal at time instant.



* Nyquist condition $F_s \geq 2f_m$

T_s : Sampling time Sec/Sample

F_s : Sampling Freq $= \frac{1}{T_s}$ Sample/Sec.

Prove ?

* Relation between discrete freq. analog freq & Sampling freq.

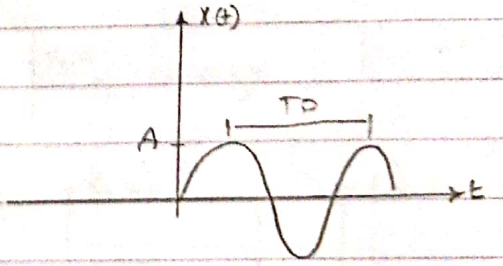
$x(t) = A\sin(2\pi F t + \theta)$

A... Amplitude

F... Frequency $= \frac{1}{T_p}$ [Hz]

T_p... Periodic time [Sec]

θ... Phase shift [°]



$x[nT_s] = A\sin(2\pi F n T_s + \theta)$

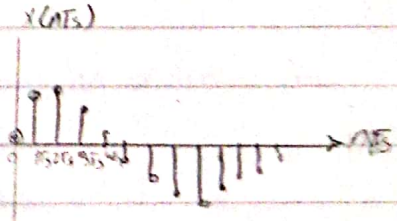
$\because T_s = \frac{1}{F_s}$

$\therefore x[nT_s] = A\sin(2\pi \frac{F}{F_s} n + \theta) = A\sin(2\pi \frac{F}{F_s} n + \theta)$

analog freq [Cycle/Sec]

$\therefore f = \frac{F}{F_s}$ ← sampling freq [Sample/Sec]

digital freq Cycle/Sample



Prove ?

* Periodicity of $x(n)$ $\Rightarrow x(n) = \cos(2\pi F n)$ Let $\theta = 0$, $A = 1$

$x(n + N_p) = \cos(2\pi F(n + N_p))$

$= \cos(2\pi F n + 2\pi F N_p)$

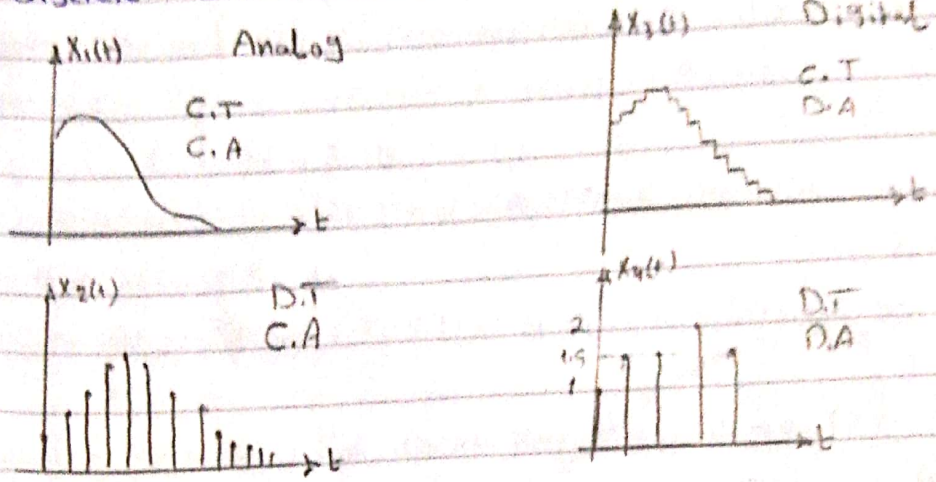
$2\pi F N_p = \text{Integer} = 2\pi K$

$F = \frac{K}{N_p}$

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3) Continous Value Signal : Signal takes its values from infinite set of possible values

vs Discrete Value Signal : Signal takes its values from finite set of possible values

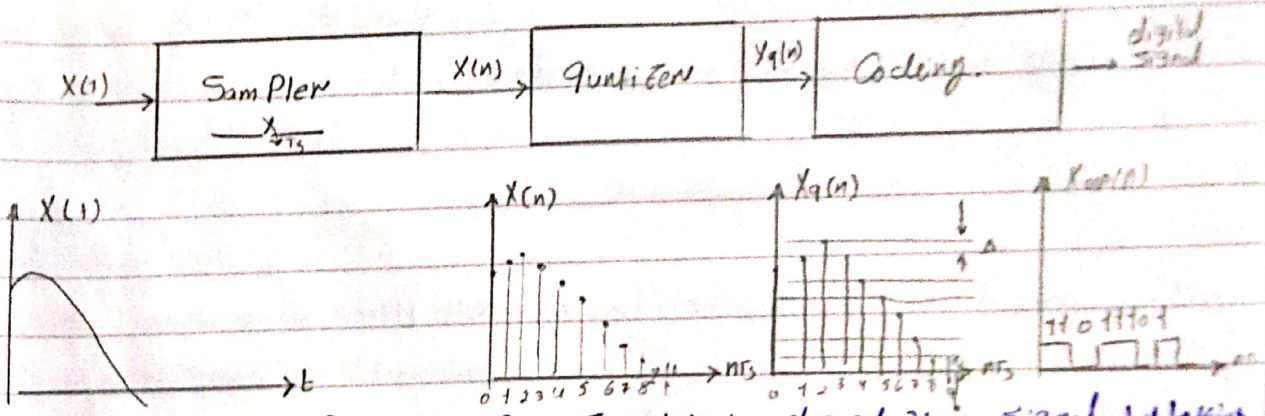


4) Random: can't be described by a function

Deterministic: Can be described by a function. EX $S_1(t) = 5t$.

---

* Basic 3-operation in A/D.



* Sampling : Conversion of continous time signal in to discret time signal by taking

Sampler $X_a(t) \Rightarrow X_a[nT_s] = X_a[n]$

* Quantization : Conversion of Continuous discret time continous value signal into discrete time discret value signal

Quantization errore : difference between unquantized and quantized signal

$$eq(n) = X_q(n) - X(n) \quad \Rightarrow -\frac{\Delta}{2} \leq eq(n) \leq \frac{\Delta}{2} \quad \Rightarrow eq(n)|_{max} = \frac{\Delta}{2}$$

Dynamic range

$$\Delta = \frac{X_{max} - X_{min}}{L-1} \qquad \frac{3-0}{3} = 1$$

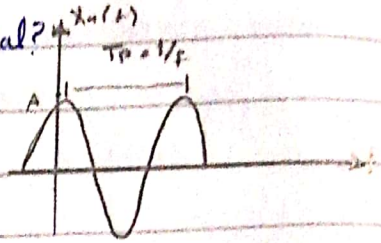
* Coding : each quantized level represented in a binary code (n-bit)

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* the limitation of Digital Signal Processing :

→ the speed of operation of A/D converter and digital signal Processor.

* What are the main Chs of Continous time sinsoidal signal?



1) every Fixed value of Frequency, F, $x_a(t)$ is Periodic

$$x_a(t+T_p) = x_a(t)$$

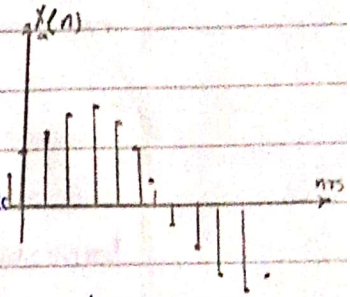
2) Continous time sinsoidal signal with diffrent Frequencies are themselves different.

$$x_a(t) = A\cos(2\pi Ft + \theta)$$

3) increasing in Frequency result in an increasing in rate of osc.

* What are the main Chs of discrete time sinsoidal signal?

1) discrete time sinsoid signal is Periodic if its Frequency F is rational number, $x_a(n)$ is Periodic with the Period $N (N>0)$ if and only if $x_a(n+N) = x_a(n)$



2) discrete time sinsoid signal with ~~operation between~~ separated by an integer Multiple of 2π are identical.

3) highest rate of oscillation in discrete time sinsoid is at $(\Omega = \pi)$ or $(n = \pi)$ $(F = \frac{1}{2})$ or $(F = -1/2)$

* Quntization :

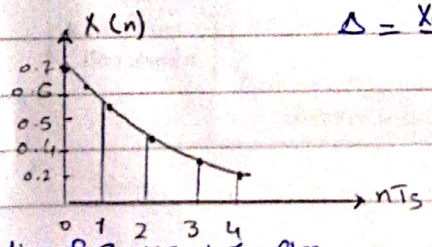
→ we can either simply discared them (trunction) or discard them by rounding the resulting number (rounding)

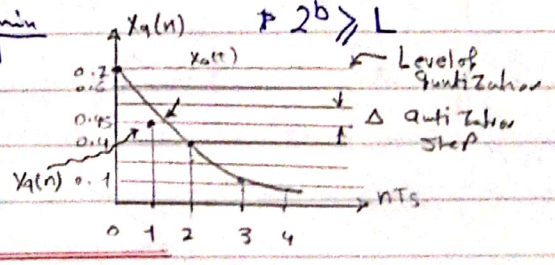
$$e_q(n) = x_q(n) - x(n)$$

$$-\frac{\Delta}{2} \leq e_q(n) \leq \frac{\Delta}{2}$$

(to minmize it inc No. of levels)

$$\Delta = \frac{x_{max} - x_{min}}{L-1}$$





$b \geq 2^b \geq L$

* Coding of quntized samples.

→ coding process in A/D converter assigns a unique binary number to each quantized level if we have 'L' levels we need atleast L different binary numbers, with a word Code length of b bits we can creat 2^b diff binary no.

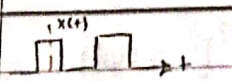
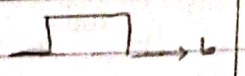
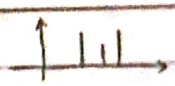
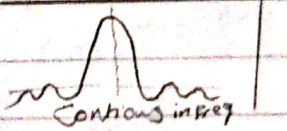
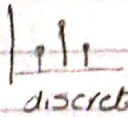
hence $2^b \geq L \Rightarrow b \geq \log_2 L$

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* Coding of Quantized Samples:

- The Coding Process in an A/D converter assigns a unique binary number to each quantization level.
- If we have "L" levels we need atleast L different binary numbers, with a word length of b bits we can create 2^b different binary numbers
  hence we have $2^b \geq L$ or equivalently $b \geq \log_2 L$

Compare between ~~FD~~ Fourier series, Fourier transform, DFT

	Fourier series	Fourier Transform	DFT
definition	used to represent continous and periodic functions as sum of Sin, Cosine waves	take signal and represent it in terms of the freq that make up that signal "Cont and Aperiodic"	used to get the spectrum of discrete & Aperiodic signal
Type of signal	Continous & Periodic	continous and Aperiodic	discrete and Aperiodic
wave form in time domain			
equations	$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega t + b_n \sin n\omega t$	$X(F) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi Ft}\, dt$	$X(K) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi nK}{N}}$ $K = 0, 1, 2, 3, \ldots N-1$
wave form in Freq. domain	Const. in Freq  Sine wave  discrete in Freq		

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Prove 3=

Prove that the sequence $X(K)$ of $X(n)$ is Periodic

$$X(K) = \sum_{n=0}^{N-1} X(n)\, e^{-\frac{j2\pi nK}{N}} \qquad K = 0, 1, 2, \dots N-1$$

$$X(K+N) = \sum_{n=0}^{N-1} X(n)\, e^{-\frac{j2\pi n}{N}(K+N)}$$

$$X(K+N) = \sum_{n=0}^{N-1} X(n)\, e^{-\frac{j2\pi nK}{N} - \frac{j2\pi nN}{N}}$$

$$X(K+N) = \sum_{n=0}^{N-1} X(n)\, e^{-\frac{j2\pi nK}{N}}\, e^{-j2\pi n}$$

$\because e^{-j2\pi n} = 1$

$$\therefore X(K+N) = \sum_{n=0}^{N-1} X(n) e^{-\frac{j2\pi nK}{N}}$$

$\because X(K) = X(K+N)$

$\therefore$ Sequence $X(K)$ is Periodic every N Samples

Prove =>

* Find the relation between Fourier transform and discrete fourier transform

$$X(j\omega) = \int_{-\infty}^{\infty} X(t)\, e^{-j\omega t}\, dt$$

$\int_{-\infty}^{\infty} \Rightarrow \sum_{n=0}^{N-1}$? $t \Rightarrow nT_s$? $X(t) \Rightarrow X(n)$? $\omega \Rightarrow K\Omega$? discrete ang. freq. $\Omega = \frac{2\pi}{NT_s}$

$dt \Rightarrow T_s$

$$\therefore X(j\omega) = \sum_{n=0}^{\infty} X(n)\, e^{-\frac{j2\pi KnT_s}{NT_s}}\, (T_s)$$

$$X(j\omega) = T_s \sum_{n=0}^{N-1} X(n)\, e^{-\frac{j2\pi Kn}{N}}$$

$$X(j\omega) = T_s\, X(K)$$

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Ques

Find relation between IFT & IDFT

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

$\int_{-\infty}^{\infty} \Rightarrow \sum_{n=0}^{N-1}$, $t \to nT_s$, $\omega \to k\Omega$, $d\omega \to \Omega = \frac{2\pi}{NT_s}$

$X(j\omega) \to X(k)$

$$\therefore x(t) = \frac{1}{2\pi} \sum_{k=0}^{N-1} X(k)\, e^{j\frac{k2\pi}{NT_s} nT_s} (\Omega)$$

$$x(t) = \frac{\Omega}{2\pi} \sum_{k=0}^{N-1} X(k)\, e^{jk\frac{n2\pi}{N}}$$

$$x(t) = \frac{1}{NT_s} \sum_{k=0}^{N-1} X(k)\, e^{j2\pi\frac{kn}{N}}$$

$$x(t) = \frac{IDFT}{T_s}$$

Ques

Prove Nuestrale

the digital frequency

$F < \frac{1}{2}$ cycle/sample

$\therefore F = \frac{f}{f_s}$

$\therefore \frac{f}{f_s} < \frac{1}{2}$

$f \leq \frac{f_s}{2} \Rightarrow f_s \geq 2f$ "Nyquist Condition"

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Chapter (3)

Definitions

* Correlation: Consider how two data sequences, each consisting of simultaneously sampled values taken from the two corresponding waveforms might be compared.

$$r_{12}(j) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n)\, x_2(n+j)$$

* End effect: As x_2 is shifted to the left the waveforms no longer overlap and data at the end of the sequences no longer form pair products.

* Correlation factor: $\rho_{12}(j)$ is known as the cross-correlation coefficient. Its value always lies between −1 and +1, +1 means 100% correlation in the same sense, −1 means 100% correlation in the opposing sense. A value of 0 signifies zero correlation.

$$\rho_{12}(j) = \frac{r_{12}(j)}{\frac{1}{N}\left[\sum_{n=0}^{N-1} x_1^2(n) \sum_{n=0}^{N-1} x_2^2(n)\right]^{1/2}}$$

* Fast Correlation: $r_{12}(j) = \frac{1}{N} F_D^{-1}\left[X_1^*(K)\, X_2(K)\right]$

F_D^{-1} ... denotes the inverse discrete Fourier transform

* convolution: it is how the input to a system interacts with the system to produce the output.

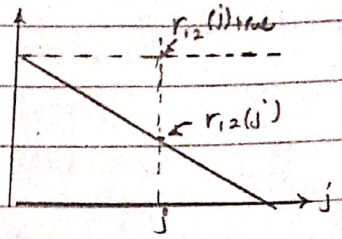
$$y(n) = \sum_{m=0}^{n} h(n-m)\, x(m)$$

$$y(n) = \sum_{m=0}^{n} h(m)\, x(n-m)$$

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* The effect of the end-effect on the cross correlation $r_{12}(j)$

① One possible solution is to make one of the sequences x_{12} twice as long as the required length for correlation. this could be achieved by recording more data, or if one of the sequences is



② if one of the sequences were periodic, by repeating the sequence

③ Another possibility is to add a correction to all computed values.

$$r_{12}(j)_{+ve} = r_{12}(j') + \frac{j'}{N} r_{12}(0)$$

Prove by conv. compact form.

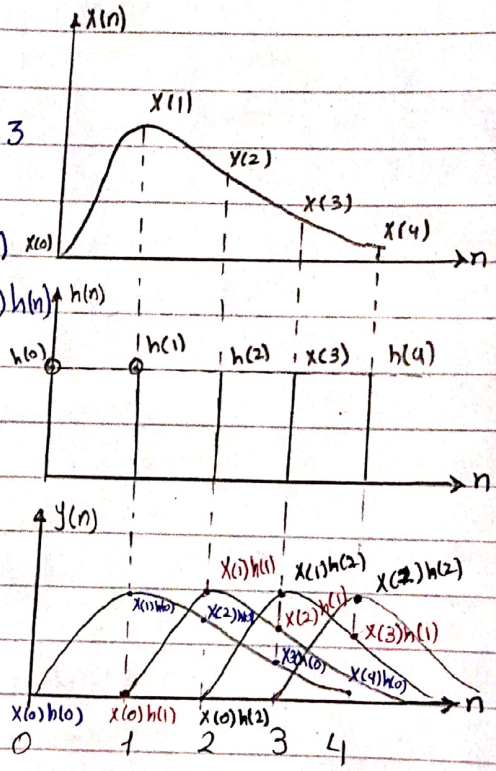
$\therefore y(0) = x(0)h(0) \rightarrow 1$

$y(1) = x(0)h(1) + x(1)h(0) \rightarrow 2$

$y(2) = x(0)h(2) + x(1)h(1) + x(2)h(0) \rightarrow 3$

$\therefore y(n) = x(0)h(n) + x(1)h(n-1) + x(2)h(n-2) + \dots + x(n)h(0)$

or $y(n) = x(n)h(0) + x(n-1)h(1) + x(n-2)h(2) + \dots + x(0)h(n)$



∴

Generally

$$y(n) = \sum_{m=0}^{n} x(m)\,h(n-m)$$

$$= \sum_{m=0}^{n} h(m)\,x(n-m)$$

Is fixed
nd Signal
tandfinds
P=0

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1] Correlation: degree of similarity between 2 signals
- auto correlation "same sig"
- cross correlation "diff sig"

① direct Method

1-a) Auto Correlation (No need for N_{new})

$$r_{11}(j) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n)\, x_1(n+j)$$

j ... lag (delay) value

N ... Number of Samples.

shift left ←

Sequences					J	$r_{11}(j)$	$\rho_{11}(j)$
$x_1(n)$	x(0)	x(1)	x(2)	x(3)	–	–	–
$x_1(n)$	x(0)	x(1)	x(2)	x(3)	0		
	x(1)	x(2)	x(3)	0	1		
$x_1(-j)$	x(2)	x(3)	0	0	2		
	x(3)	0	0	0	3		

$$\rho_{11}(j) = \frac{r_{11}(j)}{\frac{1}{N_1}\sum_{n=0}^{N-1} x_1^2(n)}$$

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1-b) Cross Correlation (Must get $N_{new} = N_1 + N_2$) زيرو بادينغ

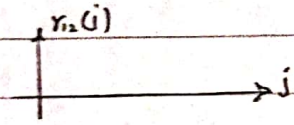
$$r_{12}(j) = \frac{1}{N_{new}} \sum_{n=0}^{N_{new}-1} x_1(n)\, x_2(n+j)$$

shift left and rotate

Sequence								J	$r_{12}(j)$	$\rho_{12}(j)$
$x_1(n)$	x(0)	–	–	–	–	–	x(7)	–	–	–
	x(1)	–	–	–	–	–	x(0)			
	x(2)	–	–	–	–	–	x(1)			
	x(7)	–	–	–	–	–	x(6)			

$$\rho_{12}(j) = \frac{r_{12}(j)}{\frac{1}{N_{new}}\left[\sum_{n=0}^{N_{new}-1} x_1^2(n) \cdot \sum_{n=0}^{N_{new}-1} x_2^2(n)\right]}$$

* graph between lag & correlation value



② Fast correlation "correlation theory method"

① conj FFT ② conj FFT ③ IFFT

$$r_{12}(j) = \frac{1}{N} F^{-1}\left[X_1^*(k) \cdot X_2(k)\right]$$

2] Convolution: Find o/p signal of system

$x(n) \rightarrow$ [System $h(n)$] $\rightarrow y(n)$

(Must get $N_{new} = N_1 + N_2 - 1$)

2-a) discrete conv

$N_2 - 1$ zeros, $N_1 - 1$ zeros, Shift right

$x(n)$			Sequence							n	y(n)
	0	0	x(0)	x(1)	x(2)	x(3)	x(4)				
$h(n)$	h(n)	h(n-1)	h(0)	0	0	0	0			0	
$h(n-1)$	0	h(n)	h(n-1)	h(0)	0	0	0			1	
	0	0	h(n)	h(n-1)	h(0)	0	0			2	
	0	0	0	h(n)	h(n-1)	h(0)	0				
	0	0	0	0	h(n)	h(n-1)	h(0)			$N-1$	

2-b) Analytic Convolution

① Make one of two signals fixed
② Find the inversion of 2nd signal
③ shift 2nd signal to right and find o/p
④ repeat stage 3 until o/p = 0

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Chapter (4)...

* definitions

* **Linear system:** A discrete time signal is said to be linear if it obeys the principles of superposition

$$a_1 x_1(n) + a_2 x_2(n) \longrightarrow a_1 y_1(n) + a_2 y_2(n)$$

where a_1, a_2 are arbitary constants.

* **Time invariant system:** A discrete time signal is said to be time invariat if it's output is independant of the time the i/p is applied

$$x(n) \longrightarrow y(n)$$

$$x(n-k) \longrightarrow y(n-k)$$

* **Causal System:** Causal D-T system has zero impulse responce before time zero

i.e.: $h(k) = 0$ for all $k < 0$

also the D-T seqvence, says, $x(n)$ is causal if it's value is zero before time zero

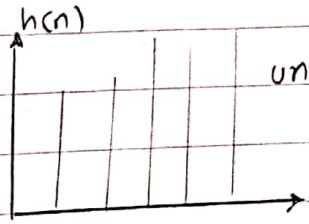
i.e.: $x(n) = 0$ for all $n < 0$

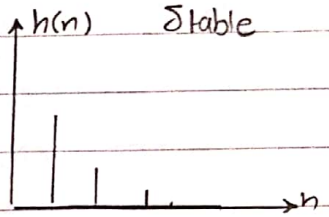
For continous time signals or sys the are causal if
$x(t) = 0$ for all $t < 0$ for signal
$h(t) = 0$ for all $t < 0$ for system

discuss2 order in

* **System Stability:** LTIS is Stable if the impulse responce $h(k)$, satisfies the following Condition:

$$\sum_{k=-\infty}^{\infty} h(k) < \infty$$





* **ROC "region of convergence":** the region where the Z-transform converges is known as region of convergence.

* **BIBO: "bounded i/p bounded o/p":** An LTI System is said to be BIBO stable if and only if it satisfies criterion

$$\sum_{k=0}^{\infty} |h(k)| < \infty$$

$h(k)$ ... impulse response of the system.

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* difference equations of discrete time systems: the difference equation specifies the actual operations that must be performed by the discrete time system on the input data, in the time domain, in order to generate the desired output

$$Y(n) = \sum_{k=0}^{N} a_k X(n-k) - \sum_{k=1}^{M} b_k Y(n-k)$$

Where $X(n)$... input samples
$Y(n)$... output samples
$Y(n-k)$... Previous output
a_k , b_k ... system coefficients.

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* Summary for questions.

Laplace → Contious time signal
Z-domain → discret time domain
Z-trans → Conv. from time domain to z domain.

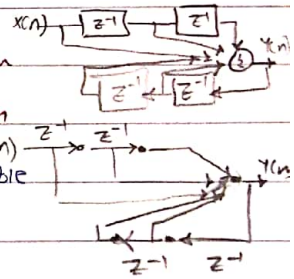
$$H(z) = \sum_{n=-\infty}^{\infty} h(n)\, z^{-n}$$

1] given transfer function $H(z)$

O Symbol ← Zeros / Poles

Req: ① Plot Pole-zero digram → ① Mul by highest-verbuer, analysthe eq.
② check Stability → ① all Poles inside circle "Stable"
② one Pole outside circle "unstable" X Symbol
③ one pole on circle "margeral Stable"
③ Find ROC ~→ all values except Poles
④ Find diff. eq. $\frac{Y(n)}{X(n)} = \frac{z^{-a}\, z^{-b}}{\cdots}$ get $Y(n)$
⑤ PLot the realization digram. → ① Block digram
→ ② node digram

Signal Flow Chart



Noted:-
① if the trans fer func has Praçet in denimnator $(\ )^2$ ∴ 2nd order Pole
② $x(n) \rightarrow \boxed{z^{-2}} \rightarrow x(n-2)$ X



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For finding h(n) samples value

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} \cdots}{b_0 + b_1 z^{-1} + b_2 z^{-2} \cdots}$$

1] Power series

a) recursive form.

$$h(n) = \left[an - \sum_{i=1}^{n} h(n-i) b_i \right] / b_0$$

$h(0) = a_0 / b_0$

b) Long division. $h(1) = 0$

$$H(z) = \frac{1 - z^{-1}}{1 - z^{-1} - 6z^{-2}}$$

$h(0)$ $h(2)$ $h(3)$

$1 + 6z^{-2} + 6z^{-3}$

$1 - z^{-1} - 6z^{-2} \;\big|\; 1 - z^{-1}$

$\ominus \; \pm z^{-1} \mp 6z^{-2}$

$6z^{-2}$

$\ominus 6z^{-2} \mp 6z^{-3} - 36z^{-4}$

$6z^{-3} + 36z^{-4}$

2] Partial fraction

$$\frac{H(z)}{z} = \frac{A}{z - \alpha} + \frac{B}{z - \alpha_2} \cdots$$

For multiple pole $\Rightarrow \frac{H(z)}{z} = \frac{A}{(z-1)} + \frac{B_1}{(z-2)^2} + \frac{B_2}{(z-2)}$

$$B_2 \big|_{z=2} = \frac{d}{dz}\left[\frac{H(z)}{z} * (z-2)^2 \right]$$

3] Residue :

$h(n) = \text{Res}[F(z), \text{Pole}_1] + \text{Res}[F(z), \text{Pole}_2] \cdots$

$F(z) = H(z)\, z^{n-1}$

For multible pole: $h(n) = \text{Res}[F(z), \text{Pole}_1] + \text{Re}[F(z), \text{Pole}_2]$ (ordinary pole / repeated pole)

$$\text{Res}[F(z), \text{Pole}_2] = \frac{d}{dz} F(z) \Big|_{z = \text{Pole}_2}$$

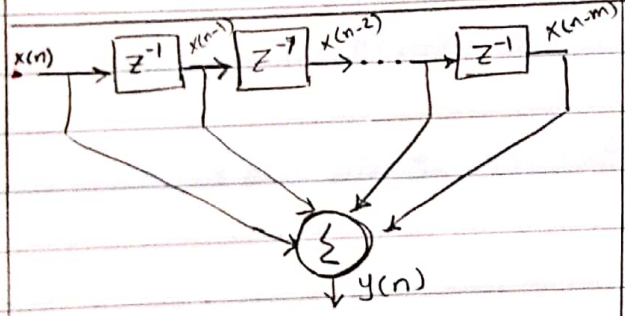
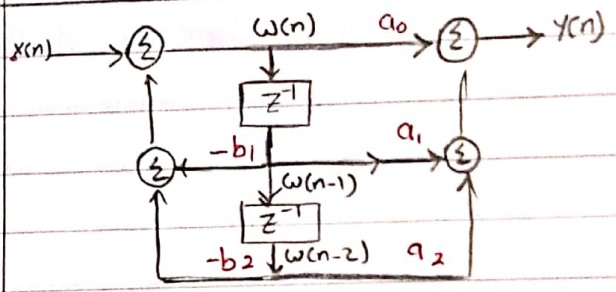
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Chapter (5)

* limitations (disadvantage) of digital filters

1- speed limitation
2- Finite word length effect
3- Long design and development time

* Compare between FIR and IIR

	FIR	IIR
	• Finite impulse response	• infinite impulse response
	• $H(z) = \sum_{k=0}^{N-1} h(k)\, z^{-k}$	• $H(z) = \dfrac{\sum_{k=0}^{N} a_k z^{-k}}{1 + \sum_{k=1}^{N} b_k z^{-k}}$
		
	• Difference eq. $y(n) = \sum_{k=0}^{m} h(k)x(n-k)$ $y(n) = h(0)x(n) + h(1)x(n-1) \dots + h(m)x(n-m)$	• Difference eq.: $w(n) = x(n) - b_1 w(n-1) - b_2 w(n-2)$ $y(n) = a_0 w(n) + a_1 w(n-1) + a_2 w(n-2)$
	• Delay ↓ • Stability : always stable "No. Feedback" • Accuracy ↓	• Delay ↑ • Stability : may stable and may not • Accuracy ↑
• Realization	transversal (direct) Frequency Sampling Fast convolution	direct form Cascade Parallel
	Lattice	

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* Choosing between FIR and IIR

1) FIR filter have linear phase response, so that no phase distortion introduced into the signal by the filter, it's important for many applications as: data transmission, biomedicine, digital audio and image processing while phase responses of IIR filters are non linear

2) FIR filter always stable, the stability of IIR filters can't always be guaranteed

3) the effect of round off noise and quantization errors are less in FIR than IIR

4) FIR requires more coefficients for sharp filters than IIR

5) Analogue filters can be easily transformed into equivalent IIR digital filter while it's not possible with FIR filters

6) FIR is algebrically more difficult to synthesize, if CAD support is not available

* Computational and storage requirments.

	FIR	IIR
No. of multiplications	12	5
No. of additions	11	4
Storage locations	24	8

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* Filters design steps

1) Specification of the filter requirments
   it include
   i) signal characteristics (type of signal source and sink, I/o interface, data rates and width, and highest frequency of intersest)
   ii) the characteristics of filter (Amplitude, Phase response, tolerance, speed)
   iii) the manner (way) of implementation
   iv) other design constrains (for ex: the cost of the filter)

2) Calculation of suitable filter coefficients
   - they are several methods of calculating filter coefficients. the most widely used
     - Impulse invariant (IIR)
     - Bilinear transformation (IIR)
     - Pole-zero Placement (IIR)
     - Window (FIR)
     - Frequency sampling (FIR)
     - optimal (FIR)

3) Representation of filter by suitable structure (realization)

FIR	IIR
transversal (direct)	direct form
Frequency sampling	Cascade
Fast convolution	Parallel
lattice	

4) Analysis of finite word length effects
   main sources of performace degradation in digital filters are.
   1- input/output signal quntization
   2- coefficient quntization
   3- arithmatic round off errors
   4- over flow

5) implemetation of a filter.

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